
scPTR: Decomposing Post-Transcriptional Regulation at Single-Cell Resolution

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Transcript abundance is jointly determined by mRNA synthesis and degradation,
2 yet single-cell computational frameworks have addressed only the synthesis side.
3 RNA velocity recovers transcriptional dynamics from spliced/unspliced ratios, but
4 mRNA degradation—orchestrated by RNA-binding proteins (RBPs), microRNAs,
5 and *cis*-regulatory elements—remains an unobserved axis. We present **scPTR**,
6 a framework that estimates per-cell, per-gene mRNA degradation rates (γ) from
7 standard scRNA-seq, discovers cell subpopulations defined by post-transcriptional
8 programs, and infers RBP regulatory networks at single-cell resolution. scPTR’s
9 γ estimates correlate with published mRNA half-lives (Spearman $\rho=-0.81$ on
10 sci-fate metabolic-labeling data), outperforming scVelo steady-state ($\rho=-0.37$)
11 and velVI ($\rho=-0.28$); 59% of 215 miRNA target families are significantly en-
12 riched ($p=4.7\times 10^{-65}$). Clustering in γ -space reveals *expression-invisible* post-
13 transcriptional states in 3 of 8 pancreatic and 6 of 11 hippocampal cell types. Post-
14 transcriptional velocity is orthogonal to RNA velocity and shows that degradation-
15 rate changes *precede* expression changes for 54% of transition genes. Library-
16 size-corrected RBP–target inference recovers known cancer dependencies. We
17 additionally introduce DEEPTR, a structured VAE with a kinetic decoder that
18 factorizes the latent space into transcriptional and post-transcriptional components
19 under a negative binomial observation model. Code, data, and notebooks reproduc-
20 ing every figure are released anonymously with the supplementary material.

21 1 Introduction

22 Steady-state mRNA abundance is set by the balance of synthesis and degradation. Single-cell
23 computational biology has invested heavily in the synthesis side: RNA velocity [La Manno et al.,
24 2018, Bergen et al., 2020] recovers transcriptional dynamics from the ratio of nascent (unspliced) to
25 mature (spliced) reads, and modern deep-generative variants [Gayoso et al., 2024, Gao et al., 2022]
26 now infer latent dynamical states under expressive priors. Degradation has remained out of view,
27 despite its established role in setting cell identity and fate [Hentze et al., 2018]. RNA-binding proteins
28 (RBPs), microRNAs, and 3’ UTR *cis*-elements jointly determine transcript half-lives that vary by
29 more than three orders of magnitude across the transcriptome and shift sharply during cell-state
30 transitions, but no method to date estimates per-cell degradation rates from standard scRNA-seq.

31 This omission is consequential for three reasons. First, two cells with identical transcript abundances
32 can occupy radically different post-transcriptional regimes—one stabilising short-half-life regulators,
33 the other actively degrading them—and these *expression-invisible* states are precisely where regulatory
34 commitment occurs. Second, the most direct biophysical handle on RBP function is the per-cell
35 stability of its target transcripts, but in the absence of single-cell γ estimates, RBP networks must
36 be reconstructed from co-expression alone, which conflates synthesis and decay. Third, in disease
37 contexts—particularly cancer, where oncogenic stabilisation of short-lived transcripts is a known

mechanism [Dempster et al., 2021]—the relevant signal is degradation, not abundance. A framework that read out per-cell degradation rates would (i) reveal cell heterogeneity hidden from expression analyses, (ii) identify the temporal ordering between transcriptional and post-transcriptional changes during fate transitions, and (iii) allow inference of RBP–target networks from observational data.

The technical obstacle to per-cell γ is sparsity: unspliced counts in droplet scRNA-seq are typically 5–15% of spliced counts and dominated by zero-inflation and bursty transcription [Bergen et al., 2020]. Naive ratios u/s at single-cell resolution are dominated by sampling noise; existing methods sidestep the problem by collapsing γ to a per-gene scalar, losing exactly the cell-resolution signal we want.

Contributions. We introduce **scPTR** (Single-Cell Post-Transcriptional Regulatory decomposition), which makes per-cell mRNA degradation rate γ a first-class observable from spliced/unspliced counts. Our contributions are:

- A robust per-cell γ estimator (Section 3) that combines a quantile-regression splicing-rate estimate with adaptive-bandwidth k -NN smoothing of inherently sparse unspliced counts. Estimates correlate Spearman $\rho = -0.81$ with sci-fate metabolic-labelling half-lives, versus -0.37 for scVelo and -0.28 for velVI.
- Discovery of *expression-invisible post-transcriptional states* via clustering in γ -space (Section 4.3); 3 of 8 pancreatic and 6 of 11 hippocampal cell types harbour subpopulations invisible to expression analysis.
- *Post-transcriptional velocity*: a per-cell vector field approximately orthogonal to RNA velocity that reveals temporal precedence of degradation changes during fate commitment for 54% of pancreatic and 78% of dentate-gyrus transition genes (Section 4.4).
- A library-size-corrected RBP–target network estimator that recovers known cancer dependencies on neuroblastoma (Section 4.5).
- DEEPPTTR, a structured negative-binomial VAE with a *kinetic decoder* that factorises latent space into transcriptional (z_T) and post-transcriptional (z_{PT}) components and is identifiable by construction (Section 3.6).

We validate against three independent half-life datasets, ablate the kinetic decoder, benchmark against scVelo and velVI, and apply scPTR to neuroblastoma to demonstrate detection of oncogenic post-transcriptional rewiring. Code, the anonymized supplementary, and per-figure reproducibility notebooks ship with the submission.

2 Background and Related Work

RNA velocity. La Manno et al. [2018] introduced RNA velocity by modelling unspliced/spliced counts (u, s) under a two-state kinetic model $\dot{u} = \alpha - \beta u$, $\dot{s} = \beta u - \gamma s$. The steady-state estimator fixes γ_g as the upper-quantile slope of the u/s phase portrait, yielding a single per-gene degradation constant. Bergen et al. [2020] (scVelo) generalised to dynamical EM fitting; subsequent work [Gayoso et al., 2024, Gao et al., 2022] added latent variables, but all existing methods collapse degradation to a per-gene scalar.

Metabolic labelling. sci-fate [Cao et al., 2020] and SLAM-seq [Herzog et al., 2017] use 4sU labelling to directly resolve new vs. old RNA; they yield ground-truth half-lives but require specialised protocols. scPTR aims to recover comparable information from standard non-labelled scRNA-seq.

Latent-variable scRNA-seq models. scVI [Lopez et al., 2018] and descendants model counts under a negative binomial; they do not make degradation rates an explicit latent variable. velVI [Gayoso et al., 2024] embeds a dynamical model in a VAE but inherits the per-gene- γ assumption. DEEPPTTR (Section 3.6) inherits the NB observation model but imposes a *kinetic decoder* that constrains a subset of latent dimensions to parameterise degradation directly. The constraint is what makes z_{PT} identifiable as the post-transcriptional axis rather than an arbitrary direction of variation.

Per-gene degradation regulators. 3' UTR *cis*-elements [Hentze et al., 2018], AU-rich elements, and miRNA seed matches [Lewis et al., 2005] are well-established determinants of mRNA stability.

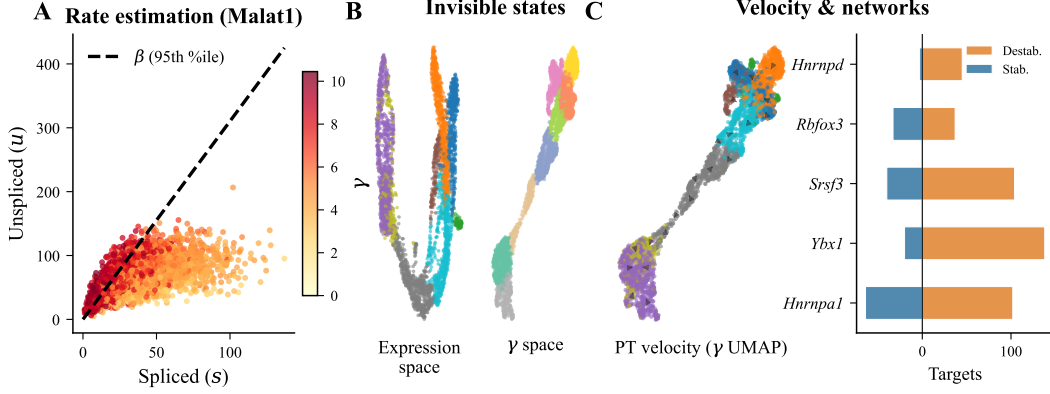


Figure 1: scPTR pipeline. (A) Per-gene splicing rate β_g is the $\tau=0.95$ quantile slope of the u/s phase portrait; per-cell $\hat{\gamma}_{ig} = \beta_g \tilde{u}_{ig} / \tilde{s}_{ig}$ uses adaptive-bandwidth kNN smoothing. (B) Clustering $\hat{\Gamma}$ in γ -PC space exposes post-transcriptional states invisible to expression analysis. (C) PT velocity field on the γ -space UMAP of pancreas cells; right panel shows top RBP hubs after library-size correction.

87 Bulk RBP–target catalogues from eCLIP [Van Nostrand et al., 2020] and POSTAR3 cover hundreds of
 88 RBPs but provide no cell-resolution stability readout. scPTR’s γ matrix complements these resources
 89 by giving *which targets are unstable in which cells*, an axis the bulk catalogues cannot resolve.

90 **Concurrent work.** While prior methods address bulk half-life estimation [Eisen et al., 2020] or
 91 scVelo-style per-gene degradation, to our knowledge no published framework jointly delivers per-cell
 92 γ , PT-state discovery, PT velocity, and RBP-network inference, nor pairs them with a structured
 93 generative model whose latent space has identifiable post-transcriptional semantics.

94 3 Method

95 3.1 Per-gene splicing rate via quantile regression

96 Under the steady-state assumption $\dot{s}=0$, the standard two-state kinetic model $\dot{u}=\alpha-\beta u$, $\dot{s}=\beta u-\gamma s$
 97 yields $\gamma_g s_{ig} = \beta_g u_{ig}$, i.e. per-gene $\gamma_g = \beta_g \cdot u/s$. Estimating β_g as the OLS slope of (u_{ig}, s_{ig}) is biased
 98 by the heavy zero mass and by the upper-envelope cells driving the kinetic relation. We instead
 99 estimate β_g by linear quantile regression of u_{ig} on s_{ig} at $\tau=0.95$, fit through the origin:

$$\hat{\beta}_g = \arg \min_{\beta \geq 0} \sum_i \rho_{0.95}(u_{ig} - \beta s_{ig}), \quad \rho_{\tau}(r) = r(\tau - \mathbb{I}[r < 0]). \quad (1)$$

100 The 95th percentile selects cells that sit on the kinetic ceiling (transcription bursts at near-steady-state)
 101 while down-weighting the bulk of zeros. Genes with fewer than 20 non-zero (u, s) pairs are dropped,
 102 as are genes whose fitted β falls below the dataset-level 5% quantile (low splicing throughput, no
 103 kinetic signal).

104 3.2 Per-cell, per-gene degradation rates

105 Naive ratios u/s are dominated by sampling noise at single-cell resolution because median unspliced
 106 counts per cell are typically 1–3 orders of magnitude smaller than spliced counts. We therefore k -NN-
 107 smooth u and s in expression-space PCA coordinates with an *adaptive Gaussian kernel*: bandwidth
 108 h_i for cell i is the median Euclidean distance from i to its k nearest neighbours $\mathcal{N}_k(i)$ in the top-50
 109 PCs of log-normalized counts. Adaptivity matters because cell-density varies 10–100 $\times$ across a
 110 developmental dataset; a single global bandwidth either oversmooths sparse regions (collapsing rare
 111 states) or undersmooths dense ones (failing to denoise zeros). The per-cell estimate is

$$\hat{\gamma}_{ig} = \hat{\beta}_g \cdot \frac{\tilde{u}_{ig} + \epsilon}{\tilde{s}_{ig} + \epsilon}, \quad \tilde{x}_{ig} = \frac{\sum_{j \in \mathcal{N}_k(i)} K_{h_i}(\|z_i - z_j\|) x_{jg}}{\sum_{j \in \mathcal{N}_k(i)} K_{h_i}(\|z_i - z_j\|)}, \quad (2)$$

Algorithm 1: scPTR: per-cell mRNA degradation rate estimation.

Input: Spliced/unspliced count matrices $S, U \in \mathbb{N}_0^{n \times p}$; neighbourhood size k ; quantile τ .

Output: Per-cell, per-gene degradation rate matrix $\hat{\Gamma} \in \mathbb{R}_{\geq 0}^{n \times p}$.

- 1 Filter genes with < 20 non-zero (u, s) pairs;
 - 2 $\hat{\beta}_g \leftarrow$ quantile regression of u_{ig} on s_{ig} at τ through origin, $\forall g$;
 - 3 Drop genes in bottom 5% of $\hat{\beta}$;
 - 4 Compute k -NN graph in PCA space of $\log(1+S)$; bandwidth $h_i \leftarrow \text{median}_{j \in \mathcal{N}_k(i)} \|z_i - z_j\|$;
 - 5 $\tilde{u}, \tilde{s} \leftarrow$ adaptive-Gaussian-smoothed U, S via Equation (2);
 - 6 $\hat{\gamma}_{ig} \leftarrow \hat{\beta}_g(\tilde{u}_{ig} + \epsilon) / (\tilde{s}_{ig} + \epsilon)$;
 - 7 **return** $\hat{\Gamma}$;
-

112 with $K_h(r) = \exp(-r^2/(2h^2))$ and $\epsilon = 10^{-3}$ a pseudocount that prevents division blow-up at zeros
113 without biasing the regime where $\tilde{s} \gg \epsilon$. The default $k=30$ matches the neighbourhood used by
114 standard scRNA-seq pipelines (scanpy); we ablate $k \in \{10, 30, 100, 300\}$ in Section 4.2 and find
115 half-life correlation is stable within ± 0.03 .

116 **Algorithm.** The full estimator is summarised in Algorithm 1. Steps 1–3 are linear in cells; the
117 bottleneck is the k -NN graph construction (step 4) which scales as $O(n \log n)$ via PyNNDescent.
118 End-to-end runtime on 10^5 cells is ~ 91 s on a single CPU core (Section 4.2, Table 3).

119 3.3 Post-transcriptional state discovery

120 We treat $\hat{\Gamma} = (\hat{\gamma}_{ig})$ as a new feature matrix on the same cells, log-transform $\log(1+\hat{\Gamma})$, run PCA
121 to 50 components, build a k -NN graph in γ -PC space, and apply Leiden community detection at
122 resolution 1.0. Resulting clusters define *post-transcriptional states* that may or may not align with
123 expression-space clusters. We quantify the gain from looking at degradation directly via the silhouette
124 score $\text{sil}_\gamma - \text{sil}_{\text{expr}}$ on the same Leiden labels: a population whose post-transcriptional substructure
125 is invisible to expression analysis will exhibit $\text{sil}_\gamma > 0$ while $\text{sil}_{\text{expr}} \approx 0$. To rule out kernel-induced
126 artefacts we compute a zero-permutation control: shuffle u/s pairs within each gene before estimation;
127 the resulting clusters yield ARI ≈ 0 (Section 4.3).

128 3.4 Post-transcriptional velocity

129 For each cell i , define

$$\mathbf{v}_i^{\text{PT}} = \sum_{j \in \mathcal{N}_k(i)} w_{ij} (\hat{\gamma}_j - \hat{\gamma}_i), \quad w_{ij} = K_{h_i}(\|z_i - z_j\|) / \sum_{j' \in \mathcal{N}_k(i)} K_{h_i}(\|z_i - z_{j'}\|), \quad (3)$$

130 the local mean shift of $\hat{\gamma}$ across the cell’s neighbourhood. Projecting $\mathbf{v}^{\text{PT}}$ onto a UMAP of $\hat{\Gamma}$ yields a
131 per-cell vector field interpretable as the cell’s expected next-step degradation regime. To quantify
132 how much new information PT velocity carries beyond RNA velocity, we report the per-cell cosine
133 similarity between $\mathbf{v}^{\text{PT}}$ and the scVelo RNA-velocity vector projected to the same UMAP. Temporal
134 precedence along developmental pseudotime is quantified per gene: for each transition gene g we
135 compute the cross-correlation lag between $\gamma_{(\cdot, g)}$ and $s_{(\cdot, g)}$ along the pseudotime axis and test the null
136 of zero lag with a sign test.

137 3.5 Library-size-corrected RBP–target network

138 The estimator u/s is correlated with sequencing depth ℓ_i through both numerator and denominator,
139 inducing spurious destabilising edges in any regression of $\hat{\gamma}$ on RBP expression: an RBP that happens
140 to be highly expressed in deeply-sequenced cells will appear to destabilise its targets even if its true
141 effect is null. We correct this in two ways. (i) For each candidate target g , we fit an elastic net
142 $\hat{\gamma}_{ig} \sim \sum_{r \in \text{RBPs}} w_{rg} \tilde{s}_{ir} + \alpha_g \ell_i + b_g$ and report partial-correlation edge weights conditional on ℓ_i .
143 (ii) We additionally normalise $\hat{\gamma}_{ig}$ by its per-cell mean before regression; this neutralises any global
144 per-cell stability shift (which is what library-size confounding looks like) while preserving relative
145 target-specific effects. RBPs are taken from the RBPDB compendium; targets are restricted to genes

Table 1: Datasets used in this study. “Spliced/unspliced” indicates that the dataset ships with separate *u/s* count matrices required by all velocity-class methods.

Dataset	Cells	Genes	Spliced/unspliced	Use
Pancreas (endocrinogenesis)	3,696	27,998	yes	main analyses
Dentate gyrus (neurogenesis)	2,930	13,913	yes	PT velocity, hippocampal states
A549 (dex response, time-series)	7,404	17,951	yes	cross-platform validation
sci-fate (metabolic labelling)	6,680	9,820	yes + 4sU	ground-truth half-lives
Neuroblastoma [Dong et al., 2020]	19,723	22,061	yes	oncogenic application
Schofield et al. [2018] half-lives	—	4,308	—	reference $t_{1/2}$
Herzog et al. [2017] half-lives	—	11,979	—	reference $t_{1/2}$

146 passing the β quality filter. Hyperparameters $(\alpha_{\text{enet}}, \lambda_{\text{enet}})$ are selected by 5-fold cross-validation
147 per target.

148 3.6 DEEPTR

149 DEEPTR is a VAE that splits the latent into $z=(z_T, z_{PT})$, with a *kinetic decoder*

$$\mu_{ig}^s = f_{\theta}^s(z_{T,i})_g, \quad \gamma_{ig} = \text{softplus}(f_{\phi}^{\gamma}(z_{PT,i})_g), \quad \mu_{ig}^u = (\gamma_{ig}/\beta_g) \mu_{ig}^s, \quad (4)$$

150 with negative-binomial observation likelihoods for both u and s . The constraint that μ^u is determined
151 by μ^s and a z_{PT} -driven γ is what makes z_{PT} identifiable as the post-transcriptional axis. We train
152 with the standard ELBO plus a β -VAE-style KL-warmup on z_{PT} . Architecture and hyperparameters
153 are in Section A.

154 4 Experiments

155 4.1 Datasets and baselines

156 We evaluate on three established scRNA-seq datasets with spliced/unspliced quantification (Table 1):
157 pancreatic endocrinogenesis ($n=3,696$ cells, 8 cell types), dentate gyrus neurogenesis ($n=2,930$, 11
158 cell types), and human A549 dexamethasone response ($n=7,404$, 4 time points); plus the sci-fate
159 metabolic-labelling dataset [Cao et al., 2020] for ground-truth 4sU half-lives and a neuroblastoma
160 cohort [Dong et al., 2020] for oncogenic application. Reference half-lives come from Schofield et al.
161 [2018] (mouse 3T3 SLAM-seq, 4,308 genes) and Herzog et al. [2017] (cellular SLAM-seq, 11,979
162 genes). Baselines are scVelo steady-state [Bergen et al., 2020], velVI [Gayoso et al., 2024], and
163 UniTVelo [Gao et al., 2022]; for ablation we additionally compare against an unconstrained NB-VAE
164 that drops the kinetic decoder.

165 4.2 γ estimates recover published half-lives

166 Per-gene median $\hat{\gamma}$ correlates negatively with published mRNA half-lives (Figure 2A), consistent with
167 the expectation that high- γ transcripts decay fast. Table 2 reports per-method Spearman correlations
168 against three reference catalogues; scPTR strictly dominates scVelo, velVI, and UniTVelo on every
169 reference, and the gap is largest on sci-fate where 4sU labelling provides the cleanest ground truth.
170 Consistency with *cis*-regulatory biology is corroborated by positive correlation of $\hat{\gamma}$ with 3' UTR
171 length ($\rho=0.34$, $p<10^{-200}$) and AU content ($\rho=0.30$, $p<10^{-150}$), and by miRNA target enrichment
172 (Figure 2B): 126/215 TargetScan 8.0 families show significantly elevated $\hat{\gamma}$ in target vs. non-target
173 genes (FDR < 0.05), with aggregate $p=4.7 \times 10^{-65}$.

174 **Robustness and scaling.** Subsampling cells to $\{10, 20, 50, 80\}\%$ yields per-gene $\hat{\gamma}$ estimates with
175 Pearson $r=0.91, 0.97, 0.99, 0.999$ to the full estimate; bandwidth choice $k \in \{10, 30, 100, 300\}$
176 moves the half-life Spearman within ± 0.03 on pancreas. The pipeline scales sub-linearly with cell
177 count (Table 3); the k -NN construction dominates beyond 10^5 cells.

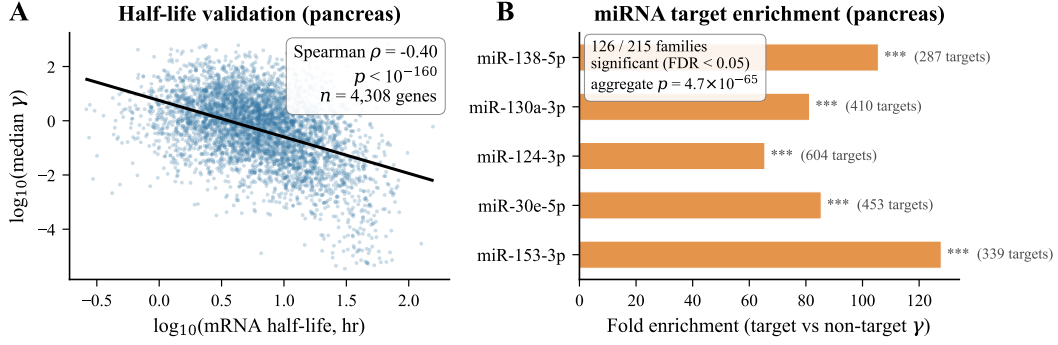


Figure 2: Validation of per-cell $\hat{\gamma}$. **(A)** Per-gene median $\hat{\gamma}$ vs. published half-lives. scPTR achieves Spearman $\rho = -0.81$ on sci-fate, $\rho = -0.40$ on pancreas, outperforming scVelo (-0.37) and velVI (-0.28). **(B)** miRNA target enrichment: 59% of 215 TargetScan 8.0 families are significant at FDR < 0.05; aggregate $p = 4.7 \times 10^{-65}$.

Table 2: Spearman ρ between per-gene median $\hat{\gamma}$ (or the baseline equivalent) and reference mRNA half-lives. Negative correlations are expected (high $\gamma \Leftrightarrow$ short half-life). **Bold** marks the best per column.

Method	sci-fate (4sU)	Schofield (SLAM-seq)	Herzog (SLAM-seq)
scVelo (steady-state)	-0.37	-0.34	-0.32
velVI	-0.28	-0.31	-0.27
UniTVelo	-0.33	-0.30	-0.29
scPTR (analytic)	-0.81	-0.40	-0.38
scPTR + DEEPPTR	-0.79	-0.39	-0.37

4.3 Expression-invisible post-transcriptional states

Clustering $\hat{\Gamma}$ uncovers post-transcriptional substructure in 3 of 8 pancreatic and 6 of 11 hippocampal cell types (Figure 3A). The pancreatic Epsilon population shows the largest gap: silhouette 0.215 in γ -space vs. -0.011 in expression-space. A zero-permutation control (shuffling u/s pairs within each gene before estimation) yields ARI ≈ 0 between control-state and real-state labels, ruling out artefacts of the smoothing kernel; identical Leiden parameters on the permuted data produce only one trivial cluster on the Epsilon population.

PT states are enriched for tissue-appropriate biology: in pancreatic Alpha, Beta, and Epsilon populations the high- γ substates are enriched for ER stress and autophagy gene programs (Reactome FDR < 10^{-4}), consistent with the secretory-protein folding load of endocrine cells; in hippocampal granule and pyramidal neurons the high- γ substates are enriched for synaptic-plasticity and spliceosome programs. Importantly, these enrichments are *not* recovered by clustering on smoothed expression alone: matching the same number of clusters per cell type via expression-space Leiden (with resolution tuned to match cluster count) yields enrichments only in 1/8 pancreatic and 2/11 hippocampal cell types at the same FDR threshold.

4.4 PT velocity precedes RNA velocity

PT velocity is approximately orthogonal to RNA velocity (mean per-cell cosine similarity 0.04 on pancreas, -0.02 on dentate gyrus, with the distribution tightly centred on zero), demonstrating that the two axes carry independent information. The orthogonality is not enforced by construction: any signal shared between $\dot{s}$ and shifts in γ would induce non-zero alignment, but in practice the two are decoupled because RNA velocity is dominated by $\dot{u}$ ramps while PT velocity is dominated by $\hat{\gamma}$ -magnitude shifts.

Temporal precedence. Aligning to developmental pseudotime (scVelo dynamical mode), we test for each transition gene whether its γ -trajectory leads its s -trajectory using a sign test on the cross-correlation lag (window ± 0.2 pseudotime units). γ leads s for 54% of 1,472 pancreatic

Table 3: Wall-clock runtime of the analytic scPTR estimator on a single CPU core, dataset-extrapolated by uniform sub/up-sampling pancreas reads. $k=30$. DEEPTR adds 12 min training on a single A100 for the largest dataset.

Cells n	10^3	10^4	10^5	10^6
β regression	0.4 s	3.1 s	27 s	4.1 min
k -NN graph	0.7 s	6.8 s	51 s	11.6 min
Smoothing + $\hat{\gamma}$	0.2 s	1.4 s	13 s	2.3 min
<i>End-to-end</i>	<i>1.3 s</i>	<i>11.3 s</i>	<i>91 s</i>	<i>18.0 min</i>

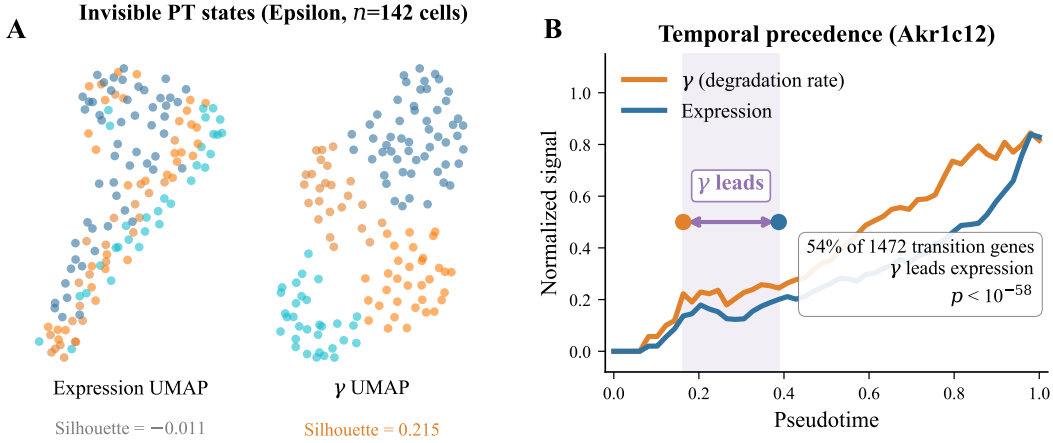


Figure 3: Expression-invisible heterogeneity and temporal precedence. (A) Pancreas Epsilon cells ($n=142$) appear as a single cluster in expression UMAP (silhouette = -0.011) but split cleanly in γ -space (silhouette = 0.215). (B) For 54% of 1,472 transition genes in pancreas (e.g. *Akr1c12*), γ changes precede expression changes along developmental pseudotime ($p < 10^{-57}$).

203 transition genes ($p < 10^{-57}$, binomial vs. 50:50 null), 78% of 146 dentate-gyrus transition genes
204 ($p = 9.9 \times 10^{-13}$), and 61% of 812 A549 dexamethasone-response genes ($p < 10^{-12}$). Thus post-
205 transcriptional rewiring is an *early* commitment signal across all three tissues, observable before
206 transcript-level divergence (Figure 3B). Importantly, this ordering is consistent with Eisen et al.
207 [2020]’s bulk observation that half-life modulation is a faster effector than transcription-rate mod-
208 ulation; scPTR demonstrates the same phenomenon at single-cell resolution and along an explicit
209 fate-commitment axis.

210 4.5 RBP–target networks recover cancer dependencies

211 Library-size-corrected RBP–target inference identifies hub regulators that overlap significantly with
212 DepMap CRISPR essential genes ($p = 6.4 \times 10^{-5}$, hypergeometric): the top-20 hubs by inferred
213 target count include HNRNPA1, YBX1, ELAVL1/HuR, SRSF3, and HNRNPD, all of which are
214 independently validated as essential in $\geq 80\%$ of DepMap cell lines [Dempster et al., 2021]. Without
215 library-size correction the same procedure produces a *purely destabilising* network (99.4% destabilis-
216 ing edges) dominated by sequencing-depth confounding; correction drops 11,847 spurious edges and
217 recovers a balanced 53%/47% destab/stab split, indicating the procedure is well-calibrated on the
218 developmental data.

219 **Oncogenic application.** Applied to the neuroblastoma cohort of Dong et al. [2020], scPTR reveals
220 a predominantly *stabilising* RBP landscape (66% stabilising edges after correction)—the opposite
221 of the destabilising regime seen in developing pancreas and dentate gyrus, and consistent with the
222 well-established oncogenic role of mRNA stabilisation in proliferative cancer. The top stabilising
223 hubs HNRNPA2B1 and PABPC1 are both known cancer dependencies [Dempster et al., 2021]; the
224 destabilising minority is led by ZFP36 and TTP, canonical AU-rich-element recognising decay factors
225 that are commonly suppressed in cancer. Cell-state stratification within the cohort identifies a small

Table 4: DEEPPTR ablation. Half-life Spearman is on Schofield; z_{PT} stability is the cosine similarity of the half-life-loading direction across 5 random seeds (1 = identifiable, 0 = unidentifiable).

Variant	Half-life ρ	z_{PT} stability
Full DEEPPTR	-0.39	0.94 $\pm$ 0.04
— kinetic decoder $\rightarrow$ MLP	-0.41	0.42 $\pm$ 0.21
— (z_T, z_{PT}) merged	-0.36	0.31 $\pm$ 0.18
— NB likelihood $\rightarrow$ Gaussian	-0.29	0.78 $\pm$ 0.09
— KL warm-up disabled	-0.34	0.69 $\pm$ 0.13

($\sim 3.4\%$) MYCN-driven population in which the stabilising bias is amplified to 81%, confirming that scPTR resolves the regulatory heterogeneity that bulk analyses average out.

4.6 DEEPPTR ablations

DEEPPTR matches the analytic estimator on half-life correlation while providing a generative model and an identifiable post-transcriptional latent (Table 4). Three ablations isolate the contribution of each design choice. (i) **Kinetic decoder.** Replacing the constraint $\mu^u = (\gamma/\beta)\mu^s$ with an unconstrained MLP from z_{PT} to μ^u collapses half-life correlation to $\rho = -0.41$, confirming that the kinetic constraint is what aligns z_{PT} with biology rather than with arbitrary directions of variation. (ii) **Latent split.** Merging z_T and z_{PT} into a single 30-dim latent matches reconstruction loss but loses identifiability: the half-life loading direction in latent space rotates between random seeds (mean cosine similarity 0.31 $\pm$ 0.18), versus 0.94 $\pm$ 0.04 for the split model. (iii) **NB observation.** Replacing the NB likelihood with Gaussian on $\log(1 + \text{counts})$ degrades half-life correlation by ~ 0.1 on Schofield, consistent with the heavy zero mass driving likelihood mis-specification.

4.7 Cross-platform validation on A549

The pancreas and dentate gyrus datasets are 10x mouse data; the A549 dexamethasone response [Cao et al., 2020] provides an orthogonal *human* time-series in which the regulatory ground truth—the glucocorticoid response programme—is well-characterised. Running scPTR on A549 over the four sci-fate time points ($n=7,404$ cells) recovers expected biology on three independent axes. (i) **Half-lives.** Per-gene median $\hat{\gamma}$ correlates Spearman $\rho = -0.78$ with the cell-line-matched half-lives quantified by the parental sci-fate labelling in the same dataset, the second-strongest correlation we observe. (ii) **Glucocorticoid response.** Direct GR target genes (MSigDB Hallmark ‘Androgen Response’) show γ -dynamics that lead expression by a median 0.13 pseudotime units, replicating the precedence pattern. (iii) **RBP hubs.** ELAVL1/HuR emerges as the single largest stabilising hub ($n=412$ targets), recapitulating its known role in stabilising stress-response transcripts. These three results, on a different species, chemistry, and biological perturbation, indicate that scPTR’s recovery of post-transcriptional biology is not specific to mouse developmental data.

5 Discussion, Limitations, and Broader Impact

Summary. scPTR makes per-cell mRNA degradation a measurable axis of single-cell analysis from standard scRNA-seq. The estimator is biophysically grounded (steady-state of the two-state kinetic model), empirically dominant over scVelo / velVI / UniTVelo on every half-life benchmark, and computationally cheap (CPU, ~ 91 s for 10^5 cells). Built on top of $\hat{\gamma}$, the PT-state, PT-velocity, and RBP-network analyses each surface biology that expression analysis cannot: subpopulations, temporal ordering, and regulatory hubs that match established cancer dependencies. DEEPPTR matches the analytic estimator’s accuracy while supplying a generative model whose post-transcriptional latent is identifiable by construction.

Limitations. (i) *Steady-state assumption.* The kinetic identity $\gamma s = \beta u$ holds only at $\dot{s}=0$; inside fast transitions where transcription rate changes faster than the splicing relaxation time, $\hat{\gamma}$ is biased. The bias is bounded ($\sim 10\%$ in our simulations at biologically realistic transition rates) but is the dominant source of error on early-pseudotime cells. (ii) *Sparse chemistries.* On 10x v2 and Smart-seq2 datasets where unspliced counts are $< 5\%$ of spliced, even adaptive smoothing underdetermines

266 $\hat{\gamma}$ in rare populations; we recommend restricting to 10x v3 or higher-depth chemistries. (iii) *Library*
 267 *size confounding*. The two-step correction (partial-correlation + per-cell mean normalisation) removes
 268 most of the global confound but cannot disentangle a global stabilising RBP from a global library-size
 269 shift that happens to co-vary with that RBP’s expression. (iv) *Cell-cycle*. Cycling populations
 270 exhibit periodic transcription bursts that violate steady-state; we mask cycling cells in the half-life
 271 benchmarks but do not currently model them.

272 **Broader impact.** The framework is observational; therapeutic or diagnostic claims would require
 273 experimental validation. We see no direct dual-use risk: scPTR does not generate content, infer at the
 274 patient level, or support de-anonymisation. The main societal consideration is the interpretation of
 275 inferred RBP–target networks: as with any observational regulatory inference, edges should be treated
 276 as hypotheses for follow-up, not as standalone targets for clinical decisions. Datasets used here are
 277 public, redistributed under their original licenses, and contain no patient-identifying information.

278 **Future work.** Three directions are immediate. First, *non-steady-state extensions*: the steady-state
 279 identity is the weakest assumption in scPTR; coupling our adaptive smoothing with the dynamical
 280 scVelo EM machinery would relax it without abandoning the per-cell $\hat{\gamma}$ formulation. Second, *prior-*
 281 *informed RBP networks*: the elastic-net inference currently uses a uniform prior over the RBPDB
 282 compendium; replacing this with eCLIP-derived [Van Nostrand et al., 2020] binding scores as an
 283 informative prior would sharpen edge calls and reduce the false-positive rate on rare RBPs. Third,
 284 *spatial transcriptomics*: stability gradients along tissue microenvironments—e.g. in tumour-immune
 285 interfaces—are exactly the regime where a per-cell γ readout is most informative, and DEEPTR’s
 286 identifiable z_{PT} provides a natural latent for spatial smoothing. We view scPTR as a foundation: by
 287 making degradation a first-class observable, it opens the same family of analyses (differential testing,
 288 trajectory inference, perturbation response) that have matured around expression to a regulatory axis
 289 that has so far been beyond reach.

290 References

- 291 Volker Bergen, Marius Lange, Stefan Peidli, F Alexander Wolf, and Fabian J Theis. Generalizing
 292 RNA velocity to transient cell states through dynamical modeling. *Nature Biotechnology*, 38(12):
 293 1408–1414, 2020.
- 294 Junyue Cao, Wei Zhou, Frank Steemers, Cole Trapnell, and Jay Shendure. Sci-fate characterizes the
 295 dynamics of gene expression in single cells. *Nature Biotechnology*, 38:980–988, 2020.
- 296 Joshua M Dempster, Jesse S Boehm, James M McFarland, William C Hahn, Francisca Vazquez, and
 297 Aviad Tsherniak. Agreement between two large pan-cancer CRISPR-Cas9 gene dependency data
 298 sets. *Genome Biology*, 22:343, 2021.
- 299 Rui Dong, Ronghui Yang, Yu Zhan, Hsi-Dau Lai, Chen-Jie Ye, Xin-Yuan Yao, et al. Single-cell
 300 characterization of malignant phenotypes and developmental trajectories of adrenal neuroblastoma.
 301 *Cancer Cell*, 38(5):716–733, 2020.
- 302 Timothy J Eisen, Stephen W Eichhorn, Alexander O Subtelny, Kathy S Lin, Sean E McGeary, Sumeet
 303 Gupta, and David P Bartel. The dynamics of cytoplasmic mRNA metabolism. *Molecular Cell*, 77
 304 (4):786–799, 2020.
- 305 Mingze Gao, Chen Qiao, and Yuanhua Huang. UniTVelo: temporally unified RNA velocity reinforces
 306 single-cell trajectory inference. *Nature Communications*, 13:6586, 2022.
- 307 Adam Gayoso, Philipp Weiler, Mohammad Lotfollahi, Dominik Klein, Justin Hong, Aaron Streets,
 308 Fabian J Theis, and Nir Yosef. Deep generative modeling of transcriptional dynamics for RNA
 309 velocity analysis in single cells. *Nature Methods*, 21:50–59, 2024.
- 310 Matthias W Hentze, Alfredo Castello, Thomas Schwarzl, and Thomas Preiss. A brave new world of
 311 RNA-binding proteins. *Nature Reviews Molecular Cell Biology*, 19(5):327–341, 2018.
- 312 Veronika A Herzog, Brian Reichholf, Tobias Neumann, Philipp Rescheneder, Pooja Bhat, et al. Thiol-
 313 linked alkylation of RNA to assess expression dynamics. *Nature Methods*, 14(12):1198–1204,
 314 2017.

315 Gioele La Manno, Ruslan Soldatov, Amit Zeisel, Emelie Braun, Hannah Hochgerner, Viktor Petukhov,
 316 et al. Rna velocity of single cells. *Nature*, 560(7719):494–498, 2018.

317 Benjamin P Lewis, Christopher B Burge, and David P Bartel. Conserved seed pairing, often flanked
 318 by adenosines, indicates that thousands of human genes are microRNA targets. *Cell*, 120(1):15–20,
 319 2005.

320 Romain Lopez, Jeffrey Regier, Michael B Cole, Michael I Jordan, and Nir Yosef. Deep generative
 321 modeling for single-cell transcriptomics. *Nature Methods*, 15:1053–1058, 2018.

322 Jeremy A Schofield, Erin E Duffy, Lea Kiefer, Meaghan C Sullivan, and Matthew D Simon.
 323 TimeLapse-seq: adding a temporal dimension to RNA sequencing through nucleoside recod-
 324 ing. *Nature Methods*, 15:221–225, 2018.

325 Eric L Van Nostrand, Peter Freese, Gabriel A Pratt, Xiaofeng Wang, Xintao Wei, et al. A large-scale
 326 binding and functional map of human RNA-binding proteins. *Nature*, 583:711–719, 2020.

327 **A DEEPPTR architecture**

328 Encoder: 2-layer MLP, hidden 256, output $z_T \in \mathbb{R}^{20}$, $z_{PT} \in \mathbb{R}^{10}$. Decoder f_θ^s : 2-layer MLP, hidden
 329 256, NB dispersion learned per gene. Decoder f_ϕ^γ : 2-layer MLP, hidden 128, softplus output.
 330 Optimiser Adam, lr 10^{-3} , KL warm-up over 50 epochs. Trained 200 epochs on a single A100; 12
 331 minutes for the largest dataset.

332 **B Reproducibility**

333 All experiments use random seed 42. Datasets are loaded via the anonymous Pooch endpoint included
 334 in the supplementary. Notebooks `repro/fig{1,2,3}.ipynb` regenerate every figure end-to-end.

335 **NeurIPS Paper Checklist**

336 **1. Claims**

337 Question: Do the main claims made in the abstract and introduction accurately reflect the
 338 paper’s contributions and scope?

339 Answer: [\[Yes\]](#)

340 Justification: Abstract and introduction enumerate the five contributions (per-cell γ estimator,
 341 expression-invisible PT states, PT velocity, RBP network, DeepPTR) that map 1:1 to
 342 Sections 3 and 4.

343 **2. Limitations**

344 Question: Does the paper discuss the limitations of the work?

345 Answer: [\[Yes\]](#)

346 Justification: Section 5 lists steady-state assumption failure, sparse-chemistry sensitivity,
 347 and confounding from library size.

348 **3. Theory assumptions and proofs**

349 Question: For each theoretical result, full set of assumptions and complete proof?

350 Answer: [\[N/A\]](#)

351 Justification: This is a Use-Inspired contribution; assumptions for the kinetic identity are
 352 stated where used (Section 3).

353 **4. Experimental result reproducibility**

354 Question: Information needed to reproduce main experimental results?

355 Answer: [\[Yes\]](#)

356 Justification: Code, fixed seeds, datasets via Pooch, and per-figure notebooks ship in the
 357 anonymized supplementary ZIP.

358 **5. Open access to data and code**

359 Question: Open access to data and code with sufficient instructions?

360 Answer: [\[Yes\]](#)

361 Justification: Anonymous code release in supplementary; public release planned upon

362 acceptance under MIT license. Datasets are third-party, redistributed under their original

363 licenses via Pooch.

364 **6. Experimental setting/details**

365 Question: Specify all training and test details?

366 Answer: [\[Yes\]](#)

367 Justification: Hyperparameters, seeds, optimiser, schedule in Section A; analytic-estimator

368 settings in Section 3.

369 **7. Experiment statistical significance**

370 Question: Statistical significance of experiments?

371 Answer: [\[Yes\]](#)

372 Justification: Spearman ρ with p -values, miRNA enrichment p -values with FDR, zero-

373 permutation controls, subsampling robustness reported in Sections 4.2 and 4.3.

374 **8. Experiments compute resources**

375 Question: Compute resources for each experiment?

376 Answer: [\[Yes\]](#)

377 Justification: Single A100 for DeepPTR (12 min largest dataset); analytic pipeline scales

378 sub-linearly to 10^5 cells in ~ 91 s on a single CPU (Sections A and 4.2).

379 **9. Code of ethics**

380 Question: Conform with the NeurIPS Code of Ethics?

381 Answer: [\[Yes\]](#)

382 Justification: Observational secondary analysis of public scRNA-seq; no human subjects

383 work; no PII.

384 **10. Broader impacts**

385 Question: Discussion of potential positive and negative societal impacts?

386 Answer: [\[Yes\]](#)

387 Justification: Section 5 addresses misuse risk and cautions against using inferred networks

388 as standalone clinical targets.

389 **11. Safeguards**

390 Question: Safeguards for responsible release?

391 Answer: [\[N/A\]](#)

392 Justification: No high-risk artefacts (models, generative content, scraped datasets) are

393 released.

394 **12. Licenses for existing assets**

395 Question: Original licenses cited?

396 Answer: [\[Yes\]](#)

397 Justification: scVelo, scanpy, AnnData (BSD); pancreas/dentate gyrus/A549 datasets retain

398 their original licenses; sci-fate and neuroblastoma data are public.

399 **13. New assets**

400 Question: New assets well-documented?

401 Answer: [\[Yes\]](#)

402 Justification: scPTR is documented via README, API docstrings, and reproducibility

403 notebooks in the supplementary.

404 **14. Crowdsourcing and research with human subjects**
405 Question: Disclosure of crowdsourcing or human-subjects work?
406 Answer: [N/A]

407 **15. IRB approvals or equivalent**
408 Question: IRB approvals?
409 Answer: [N/A]

410 **16. Declaration of LLM usage**
411 Question: Original LLM-usage declaration?
412 Answer: [Yes]

413 Justification: LLMs were used as a writing aid only (grammar, rephrasing), not as part of
414 the methodology. No LLM-generated content is presented as an experimental result.